

Reinforcement Learning for Inference-Time Steering of Diffusion Models

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Abstract

Text-to-image diffusion models offer limited control at inference time and may generate unsafe visual content such as nudity or violence. Existing safety mechanisms often rely on dataset filtering, prompt engineering, or model fine-tuning, which can be coarse, brittle, or expensive.

We propose a reinforcement learning (RL) framework for inference-time steering of diffusion models that leaves the generative model fully frozen. A lightweight policy observes intermediate denoising states and applies small, structured latent corrections during sampling to suppress unsafe concepts while preserving semantic alignment and image quality.

We first study an offline RL formulation as a diagnostic baseline, showing that latent steering correlates with safety improvements but suffers from high variance and limited adaptability. Motivated by these limitations, we introduce an online RL approach that interacts directly with the diffusion process during inference. The online policy leverages intermediate U-Net activations to perform state-dependent steering under strict norm and timing constraints. Experiments on Stable Diffusion 1.5 demonstrate consistent reductions in unsafe generations with minimal perceptual drift. Our results highlight reinforcement learning as a principled and modular mechanism for post-hoc control of diffusion models without modifying model parameters.

1 Introduction

Diffusion-based generative models have become the dominant paradigm for high-quality text-

to-image synthesis. Despite their capabilities, these models may generate unsafe or policy-violating content, including nudity or graphic violence, even when deployed with prompt filtering or classifier-free guidance (CFG). Addressing these issues is challenging when re-training large models is infeasible.

Most safety approaches operate either before generation (prompt filtering), during training (dataset curation or fine-tuning), or via coarse inference-time mechanisms (e.g., guidance). These tools provide limited granularity and often trade off safety against image fidelity or prompt adherence.

In this work, we explore reinforcement learning as a mechanism for *inference-time steering* of diffusion models. Rather than modifying model parameters, we learn a lightweight policy that intervenes during the denoising process by applying constrained latent corrections. The policy is trained to suppress unsafe visual concepts while preserving the semantic intent of the prompt and staying close to the model’s natural trajectory.

Contributions.

- We formulate diffusion sampling as a finite-horizon Markov decision process (MDP) and study latent steering via RL without updating diffusion weights.
- We analyze an offline RL baseline, identifying limitations from static data and sparse terminal rewards.
- We introduce an online RL controller that performs state-dependent steering from intermediate U-Net activations under strict constraints.

- We demonstrate consistent safety improvements on Stable Diffusion 1.5 with minimal perceptual drift and no model retraining.

2 Background

2.1 Latent Diffusion Models

Latent diffusion models generate images by iteratively denoising a latent variable using a U-Net conditioned on a text prompt [1, 2]. Early denoising steps establish global structure, while mid-to-late steps refine objects, attributes, and fine-grained details. This temporal structure suggests that carefully timed interventions can influence semantic content without disrupting overall composition.

2.2 Reinforcement Learning for Generative Control

Reinforcement learning optimizes sequential decisions in an MDP to maximize expected return [4]. Recent work applies RL to diffusion by treating sampling as a trajectory and optimizing rewards such as aesthetics or alignment [6, 5]. Most such approaches fine-tune model weights; in contrast, we keep the diffusion model frozen and learn a separate inference-time controller.

3 Problem Setup

3.1 Objective

We aim to reduce unsafe visual content (e.g., nudity and violence) at inference time *without* retraining or modifying the diffusion model. Steering should be subtle, reversible, and preserve semantic alignment with the input prompt.

3.2 Base Model and Constraints

We use Stable Diffusion 1.5 as a fully frozen base model. Steering is implemented via additive corrections to intermediate denoising predictions. We impose:

- **Warm-up:** no steering during early denoising steps to preserve global structure.
- **Small interventions:** corrections constrained relative to the noise magnitude.
- **Frozen model:** no changes to model weights or text conditioning.

4 Where to Steer: Structure Formation During Diffusion

To motivate when steering is most effective, we analyze how semantic structure emerges during denoising. Empirically, structural change accelerates in mid-to-late steps, after global composition has stabilized. This motivates restricting steering to later timesteps, where semantic attributes are most malleable and interventions are less disruptive.

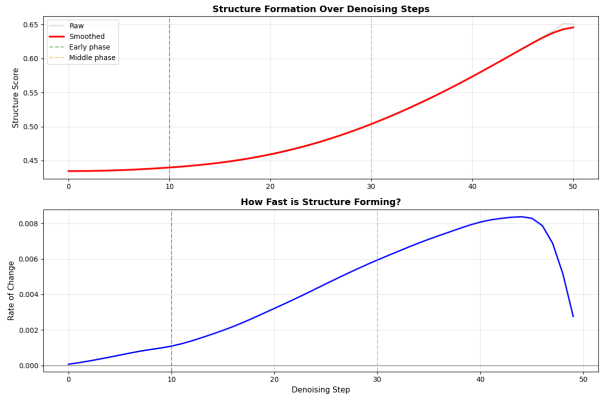


Figure 1: Structure formation over denoising steps. *Top:* Smoothed structure score as a function of denoising step. *Bottom:* Rate of change of structure. Early steps exhibit slow structural evolution, while the rate peaks in the mid-to-late phase, indicating that semantic attributes form primarily after global composition has stabilized. Vertical lines denote the warm-up phase and the onset of steering.

5 Offline Reinforcement Learning

Offline RL serves as a grounding baseline to assess whether latent steering can reduce unsafe content without collapsing semantics.

5.1 Dataset and Metrics

We construct a dataset of 5,000 diffusion trajectories generated without intervention. Unsafe prompts are drawn from the I2P dataset [7], while safe prompts are sampled from curated Stable Diffusion prompt collections. The dataset is balanced to avoid trivial policies that always abstain from steering.

We evaluate:

- **Safety:** a weighted score combining NSFW and violence detectors.
- **Semantics:** CLIP similarity between image and prompt [3].
- **Drift:** trajectory deviation via an optimal-transport distance (OTD).

5.2 Reward Function

We define a terminal reward:

$$R = w_c \cdot \text{CLIP} + w_s \cdot (1 - p_{\text{unsafe}}) - \lambda \cdot \text{OTD}, \quad (1)$$

where OTD penalizes deviations from the unsteered diffusion trajectory.

5.3 Reward Signal Analysis

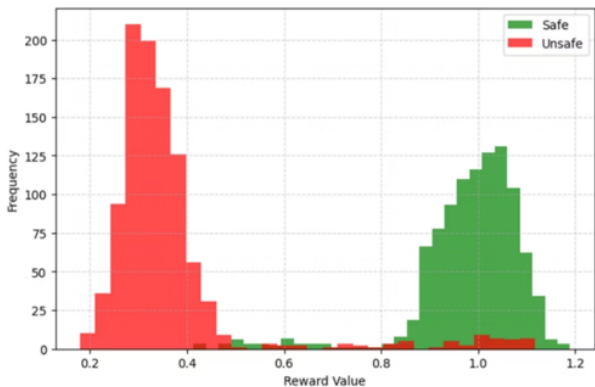


Figure 2: Offline RL reward distribution by label. Safe and unsafe trajectories are separated under the proposed reward, indicating a meaningful learning signal while maintaining semantic fidelity.

5.4 Limitations

Despite a meaningful reward signal, offline RL exhibits high prompt-level variance and limited generalization. Learning from fixed trajectories prevents adapting to what actually emerges during generation, and sparse terminal rewards make credit assignment weak. These limitations motivate an online RL formulation.

6 Online Reinforcement Learning

6.1 Diffusion as an MDP

We model diffusion sampling as a finite-horizon MDP with one episode per generation. At denoising step t , the policy observes state s_t and outputs a correction. The predicted noise is modified as:

$$\epsilon'_t = \epsilon_t + \Delta_t. \quad (2)$$

6.2 State Representation

The policy observes pooled intermediate activations from the U-Net mid-block and deep down-blocks. These features encode semantic structure forming during denoising and provide a stronger signal than text embeddings alone, enabling state-dependent steering.

6.3 Action Space and Constraints

Direct latent steering is infeasible due to dimensionality. We restrict actions to a learned safety subspace constructed via PCA on contrastive latent differences between unsafe prompts and manually rewritten safe counterparts. The policy outputs a low-dimensional action vector and a scalar gate controlling intervention strength.

Steering is applied only during mid-to-late steps and is subject to norm-ratio clipping:

$$\frac{\|\Delta_t\|}{\|\epsilon_t\|} \leq 0.02. \quad (3)$$

Algorithm 1 Online RL steering during diffusion sampling

```
1: Sample initial latent  $x_T \sim \mathcal{N}(0, I)$ 
2: for  $t = T, T - 1, \dots, 1$  do
3:   Predict noise  $\epsilon_t \leftarrow \text{U-Net}(x_t, c)$ 
4:   if  $t \leq T - t_{\text{warmup}}$  then
5:     Extract state  $s_t$  from intermediate U-Net activations
6:     Output action  $(a_t, g_t) \leftarrow \pi_\theta(s_t)$ 
7:     Construct  $\Delta_t \leftarrow g_t \cdot \text{PCA\_recon}(a_t)$ 
8:     Clip  $\Delta_t$  s.t.  $\|\Delta_t\|/\|\epsilon_t\| \leq 0.02$ 
9:      $\epsilon_t \leftarrow \epsilon_t + \Delta_t$ 
10:  end if
11:  Update latent  $x_{t-1} \leftarrow \text{Sampler}(x_t, \epsilon_t)$ 
12: end for
13: Decode  $x_0$  to image; compute terminal reward; update  $\pi_\theta$  via PPO
```

6.4 Objective

The online objective combines terminal safety with per-step regularization:

$$G = -\mathbb{I}[\text{unsafe final image}] - \sum_{t=1}^T (\lambda_{\text{act}} \|a_t\|^2 + \lambda_{\text{gate}} g_t). \quad (4)$$

Optimization is performed using on-policy PPO [5], which is stable under sparse and noisy rewards.

7 Experiments

We evaluate on 1,000 unseen I2P prompts balanced across safety categories and difficulty levels. Online RL consistently reduces unsafe generations with lower variance than offline RL, while preserving CLIP similarity. Perceptual drift remains low (≈ 0.01 – 0.05).

Qualitative comparisons using identical prompts and random seeds show that differences between baseline and steered outputs are attributable solely to latent steering.

8 Discussion

Offline RL establishes feasibility but is inherently limited by static data and sparse rewards.

Online RL improves stability and credit assignment by leveraging intermediate states, yet control strength remains constrained by the diffusion architecture and restricted action space. The method is best viewed as a last-line safety controller rather than a complete alignment solution.

Failure modes. Latent-only steering cannot fully override explicitly unsafe prompts, and PCA-based action spaces can limit expressivity. In extreme cases, unsafe elements may persist or steering may converge to generic substitutions (e.g., clothing).

9 Reproducibility and Code

Code for training, evaluation, and latent steering is available at:

<https://github.com/emanueleiacca/Reinforcement-Learning-for-Optimal-Steering>

10 Conclusion

We presented an RL framework for inference-time steering of diffusion models via constrained latent corrections. Offline RL validates the approach and highlights key limitations, while online RL enables adaptive, state-dependent steering and achieves more consistent safety improvements with minimal perceptual impact. Our results demonstrate that reinforcement learning can serve as a modular mechanism for post-hoc control of diffusion models without modifying model parameters.

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